	Extranuclear Structure	Nucleus	
Fundamental Building Blocks	Electrons	Neutrons Protons	
Field of Force	Electric Field	Electric Field	Nuclear Field
Particles emitted when changes in state take place	Photons	Photons	Electrons Positrons Neutrinos

This first horizontal row contains the fundamental building blocks, while the second one shows the field acting between them, and in the third one we find the particles emitted in the occasional changes of state.

We have still to inquire into the nature of the force which binds these building blocks of the nucleus together. It would seem logical to assume that this field, like the one within the extranuclear structure, is electrical. It is, however, easy to show that electric forces alone would not be sufficient to provide an explanation of nuclear cohesion, for the simple reason that the strongest electric effects are due to the charges carried. by the protons, and these are forces of repulsion. Therefore, a further field of another type must be operative in the nucleus. For lack of a more exact knowledge of the nature of this field, let us first give it a name, and call it *nuclear field.* Of course, in addition to it, there exists also an electric field in the nuclei, since the protons carry electric charges.

In the extranuclear structure of the atom, changes in state are accompanied by the production of particles—photons—out of the energy of the electric field. In considering the nucleus, we shall, analogously, look for particles which come into being, in conjunction with changes in state, out of the energy of its two fields, and which occasionally—in excited states—detach themselves and are emitted. Again, photons alone can correspond to the electric field; in fact, it has been discussed that in nuclear transmutations gamma-ray photons are frequently emitted, which are simply photons of an extremely short wavelength. It is obvious, therefore, that the nuclear field must be correlated with the other particles which are emitted in nuclear transformations after the lapse of a certain period of time only—namely, the electrons, positrons and neutrinos. This analogy between the extranuclear structure of the atom and its nucleus gives us a relatively simple picture of the latter.

A nucleus is built of protons and neutrons. Its building blocks interact